

Biology All In One (BAIO) Learning System

PU-II Biology · Narendra Sir – M.Sc (Biochemistry), B.Ed, KSET

Chapter 9: Biotechnology – Principles and Processes

Worksheet · Cloning Vectors for Plants/Animals & Competent Hosts

I. Multiple Choice Questions

- The Ti plasmid is associated with which organism?
(A) *Agrobacterium tumefaciens* (B) *Escherichia coli*
(C) *Bacillus subtilis* (D) *Thermus aquaticus*
- Agrobacterium tumefaciens* is naturally a:
(A) Human pathogen (B) Plant pathogen causing tumours
(C) Fungal pathogen (D) Non-pathogenic soil bacterium only
- To use the Ti plasmid as a cloning vector, scientists must:
(A) Add pathogenicity genes (B) Remove the genes responsible for pathogenicity
(C) Increase its copy number only (D) Convert it into a bacteriophage
- Retroviruses, which infect animals, are known to cause:
(A) Immunity (B) Cancers
(C) Antibiotic resistance (D) Photosynthesis
- A 'competent host' refers to a cell that:
(A) Cannot take up any DNA (B) Has the ability to take up vector DNA (recombinant DNA)
(C) Is resistant to all antibiotics (D) Lacks a cell membrane

II. Fill in the Blanks

- The Ti plasmid used as a plant cloning vector comes from _____.
- Agrobacterium tumefaciens* is naturally a plant _____ that causes tumours.
- To use the Ti plasmid as a vector, the genes responsible for _____ are removed.
- A bacterial cell capable of taking up vector DNA is called a _____ host.
- Bacterial cells are made competent by treatment with divalent cations such as _____.
- Bombardment of cells with high-velocity gold or tungsten particles is called _____.

III. Match the Following

Column A	Column B
1. Retrovirus	a. Direct injection of DNA into a nucleus
2. Microinjection	b. Disarmed animal-cell vector
3. Biolistics	c. Brief exposure to about 42°C
4. Heat shock	d. High-velocity particle bombardment

IV. True / False

- The Ti plasmid can be used directly as a vector without any modification. [True / False]

2. Retroviruses infect animals and can cause cancers. [True / False]
3. Competent host cells are treated with Ca^{2+} ions to increase membrane permeability. [True / False]
4. Biolistics is also known as the gene gun method. [True / False]
5. Microinjection is exclusively used to transform bacterial cells. [True / False]

V. 2 Marks Questions

1. Explain what it means to 'disarm' the Ti plasmid before using it as a vector.

2. Describe the CaCl_2 /heat-shock method of making bacterial cells competent.

VI. 3 Marks Questions

1. Differentiate between microinjection and biolistics as gene-delivery methods.

2. Why can retroviruses be used as cloning vectors for animal cells?

VII. 5 Marks Questions

1. Describe any two methods used to introduce recombinant DNA into a competent host cell, explaining the underlying principle of each method.
